Quantification: 543-2
==========
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 12 |
| interior lacuna count | 12 |
| frame-edge lacuna count | 0 |

Interior lacunae only: mean, median and sample SD (n = 12)
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| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2027.92 | 2066 | 537.141 | px^2 |
| major_axis_length_px | lacuna shape | 81.5016 | 86.4661 | 16.9465 | px |
| minor_axis_length_px | lacuna shape | 33.8846 | 32.3799 | 6.9212 | px |
| aspect_ratio | lacuna shape | 2.4727 | 2.3986 | 0.6666 | unitless |
| eccentricity | lacuna shape | 0.8942 | 0.9087 | 0.0558 | unitless |
| solidity | lacuna shape | 0.8887 | 0.9115 | 0.0668 | unitless |
| roots_count | headline | 9 | 9 | 2.6629 | count |
| ring_length_r30_px | headline | 410.167 | 386.5 | 87.3632 | px |
| ring_length_r60_px | reported | 1111.25 | 1113.5 | 218.065 | px |
| owned_length_px | ownership-dependent | 3601.47 | 3557.2 | 1502.1 | px |
| edge_count | ownership-dependent | 169.167 | 165 | 69.2017 | count |
| mean_edge_length_px | ownership-dependent | 21.1495 | 21.0104 | 1.7194 | px |
| perimeter_px | normalised | 222.101 | 229.693 | 45.067 | px |
| ring_area_r30_px2 | normalised | 8742.17 | 8905.5 | 1127.78 | px^2 |
| ring_area_r60_px2 | normalised | 22189.9 | 23189 | 2825.36 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9965 | 1 | 0.0123 | unitless |
| in_frame_fraction_r60 | normalised | 0.9814 | 1 | 0.0596 | unitless |
| ring_density_r30 | normalised | 0.046754 | 0.0458883 | 0.00611617 | px^-1 |
| ring_density_r60 | normalised | 0.0498661 | 0.051119 | 0.00574932 | px^-1 |
| roots_per_100px_perimeter | normalised | 4.0113 | 3.9619 | 0.6761 | per 100 px |
| ring_attached_length_r30_px | network v2 | 367.417 | 351 | 89.7962 | px |
| ring_attached_length_r60_px | network v2 | 1006.42 | 973.5 | 223.181 | px |
| ring_length_w_r30_px | network v2 | 463.474 | 438.055 | 101.083 | px |
| ring_length_w_r60_px | network v2 | 1269.36 | 1267.17 | 250.919 | px |
| sholl_crossings_r10 | network v2 | 10.8333 | 11 | 2.6572 | count |
| sholl_crossings_r20 | network v2 | 13.75 | 12.5 | 3.0189 | count |
| sholl_crossings_r30 | network v2 | 14.9167 | 15 | 3.4234 | count |
| loop_count | loops, report only | 19.9167 | 18.5 | 10.1843 | count |
| loop_edge_share | loops, report only | 0.4407 | 0.4525 | 0.0885 | share |
| ring_width_mean_r30_px | width | 7.3596 | 7.2923 | 0.3045 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02424e+06 | px^2 |
| total_skeleton_length_px | 44739 | px |
| canalicular_length_density_per_px | 0.0436802 | px^-1 |
| junction_count | 1155 | count |
| junction_density_per_px2 | 0.00112766 | px^-2 |
| lacuna_count | 12 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.172e-05 | px^-2 |
| skeleton_component_count | 148 | count |
| mean_component_length_px | 302.291 | px |
| median_component_length_px | 19.5 | px |
| field_length_density_w_per_px | 0.0497465 | px^-1 |
| field_density_without_flagged_per_px | 0.0436802 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 416 | px |
| unattached_component_count | 24 | count |
| canalicular_length_density_attached_per_px | 0.0436802 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0440863 | px^-1 |
| loop_count | 285 | count |
| loop_edge_share | 0.4985 | share |
| edge_count | 2297 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.9928 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 44739 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2494 | 90.6112 | 36.7713 | 2.4642 | 0.914 | 0.9096 | 0.4535 | 88.1403 | 271.181 | False |
| 2 | 3004 | 86.0791 | 52.8366 | 1.6292 | 0.7895 | 0.772 | 0.3181 | 207.791 | 919.033 | False |
| 3 | 1698 | 63.4601 | 34.3535 | 1.8473 | 0.8408 | 0.955 | 0.5365 | 257.425 | 340.805 | False |
| 4 | 1631 | 87.5017 | 30.956 | 2.8266 | 0.9353 | 0.741 | 0.0248 | 297.032 | 38.7695 | False |
| 5 | 1204 | 55.3793 | 28.6478 | 1.9331 | 0.8558 | 0.9177 | 0.1008 | 394.263 | 364.613 | False |
| 6 | 2066 | 81.299 | 34.8468 | 2.333 | 0.9035 | 0.8619 | 0.1545 | 472.247 | 745.532 | False |
| 7 | 1233 | 57.0332 | 27.8479 | 2.048 | 0.8727 | 0.9278 | 0.052 | 459.387 | 484.83 | False |
| 8 | 2066 | 100.448 | 26.9826 | 3.7227 | 0.9632 | 0.9018 | 0.236 | 550.262 | 877.61 | False |
| 9 | 2656 | 108.838 | 32.5449 | 3.3442 | 0.9542 | 0.8877 | -0.0333 | 642.811 | 193.327 | False |
| 10 | 2113 | 91.9731 | 30.2288 | 3.0426 | 0.9444 | 0.9135 | 0.0369 | 728.295 | 633.991 | False |
| 11 | 2173 | 86.8531 | 32.2149 | 2.6961 | 0.9287 | 0.9358 | 0.0768 | 787.166 | 480.067 | False |
| 12 | 1997 | 68.5432 | 38.3841 | 1.7857 | 0.8285 | 0.9402 | 0.1037 | 859.482 | 288.042 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 13 | 514 | 1397 | 4244.9 | 209 | 20.3105 | 7.8391 |
| 2 | 12 | 614 | 1493 | 5619.75 | 273 | 20.5852 | 7.0213 |
| 3 | 8 | 386 | 977 | 2306.85 | 104 | 22.1812 | 7.2817 |
| 4 | 9 | 382 | 710 | 1223.9 | 71 | 17.2381 | 7.0693 |
| 5 | 6 | 355 | 961 | 2569.59 | 127 | 20.233 | 7.1713 |
| 6 | 10 | 354 | 1023 | 1830.28 | 88 | 20.7986 | 7.4527 |
| 7 | 5 | 301 | 942 | 3038.05 | 124 | 24.5004 | 7.1774 |

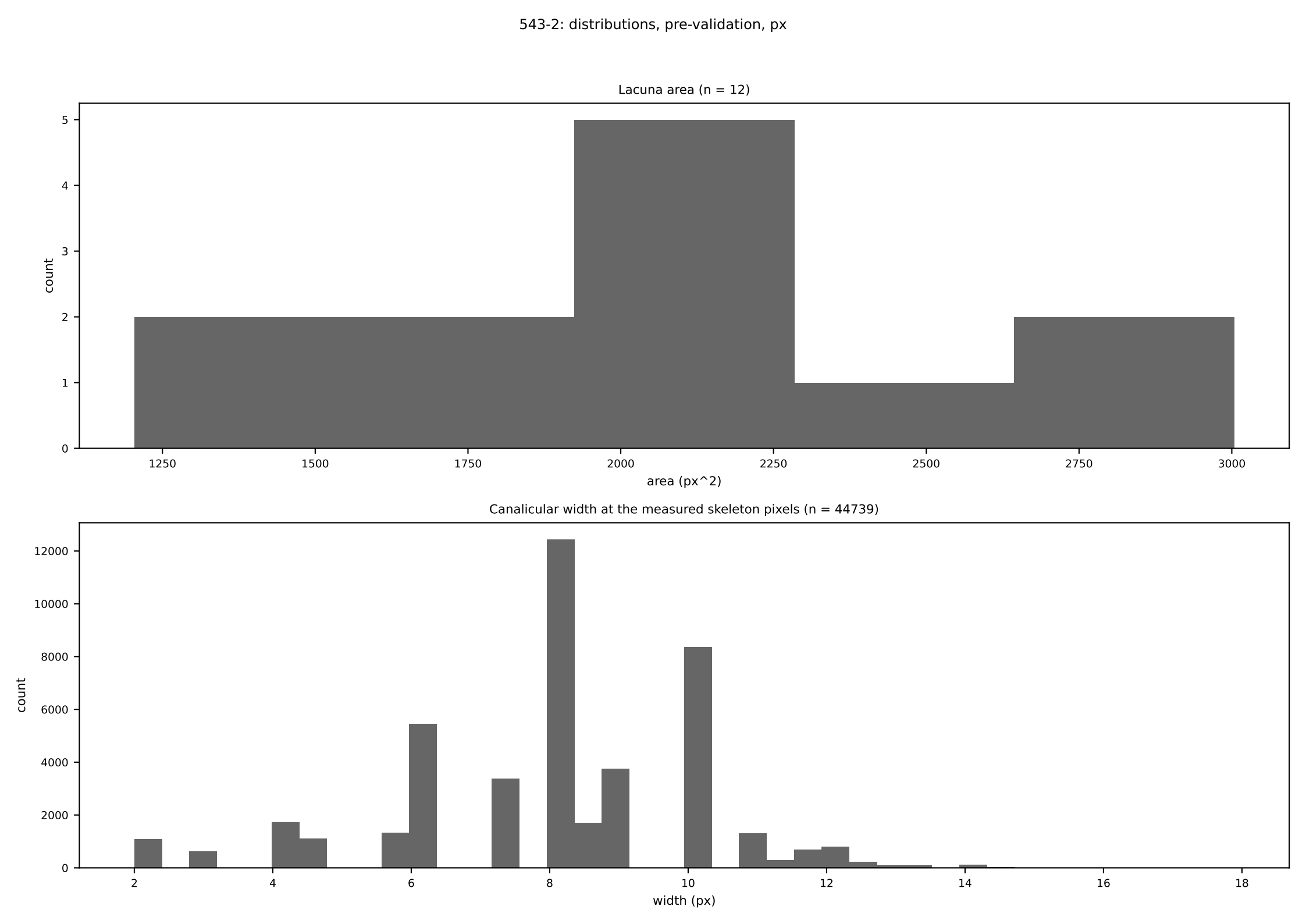
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | 12 | 454 | 1211 | 4076.35 | 190 | 21.4545 | 7.5322 |
| 9 | 11 | 424 | 1229 | 5814.11 | 266 | 21.8576 | 7.4156 |
| 10 | 9 | 387 | 1204 | 4944.79 | 233 | 21.2223 | 7.9823 |
| 11 | 7 | 435 | 1205 | 4640.02 | 205 | 22.6343 | 7.0697 |
| 12 | 6 | 316 | 983 | 2909.04 | 140 | 20.7789 | 7.3029 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 244.492 | 9510 | 23791 | 1 | 0.9843 | 0.0540484 | 0.0587197 | 5.3171 |
| 2 | 295.12 | 10519 | 26271 | 1 | 1 | 0.0583706 | 0.0568307 | 4.0661 |
| 3 | 172.894 | 7548 | 18977 | 1 | 1 | 0.0511394 | 0.0514834 | 4.6271 |
| 4 | 232.551 | 8756 | 18925 | 0.9575 | 0.7928 | 0.0436272 | 0.0375165 | 3.8701 |
| 5 | 154.267 | 7010 | 17818 | 1 | 1 | 0.0506419 | 0.0539342 | 3.8894 |
| 6 | 231.764 | 9080 | 22988 | 1 | 1 | 0.0389868 | 0.0445015 | 4.3147 |
| 7 | 155.924 | 7017 | 19168 | 1 | 1 | 0.0428958 | 0.0491444 | 3.2067 |
| 8 | 254.25 | 9429 | 23720 | 1 | 1 | 0.0481493 | 0.051054 | 4.7198 |
| 9 | 277.078 | 10094 | 25807 | 1 | 1 | 0.0420051 | 0.0476227 | 3.97 |
| 10 | 227.622 | 8939 | 23523 | 1 | 1 | 0.0432934 | 0.0511839 | 3.9539 |
| 11 | 225.522 | 8872 | 23390 | 1 | 1 | 0.0490307 | 0.0515177 | 3.1039 |
| 12 | 193.723 | 8132 | 21901 | 1 | 1 | 0.0388588 | 0.0448838 | 3.0972 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 485 | 1329 | 587.588 | 1609.02 | 13 | 16 | 16 |
| 2 | 566 | 1438 | 696.63 | 1702.3 | 15 | 21 | 21 |
| 3 | 354 | 803 | 436.505 | 1121.52 | 12 | 14 | 14 |
| 4 | 356 | 647 | 429.534 | 793.955 | 12 | 13 | 8 |
| 5 | 276 | 896 | 396.836 | 1111.18 | 8 | 11 | 15 |
| 6 | 335 | 1001 | 403.706 | 1172.76 | 10 | 11 | 11 |
| 7 | 251 | 835 | 338.108 | 1073.26 | 7 | 11 | 12 |
| 8 | 437 | 1067 | 523.446 | 1385.47 | 14 | 12 | 17 |
| 9 | 348 | 893 | 469.877 | 1407.8 | 9 | 16 | 15 |
| 10 | 306 | 1152 | 439.605 | 1362.77 | 8 | 12 | 18 |
| 11 | 382 | 1070 | 487.09 | 1361.57 | 13 | 16 | 17 |
| 12 | 313 | 946 | 352.764 | 1130.76 | 9 | 12 | 15 |


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 44739)
count
width (px)
0
2000
4000
6000
8000
10000
12000
2
4
6
8
10
12
14
16
18